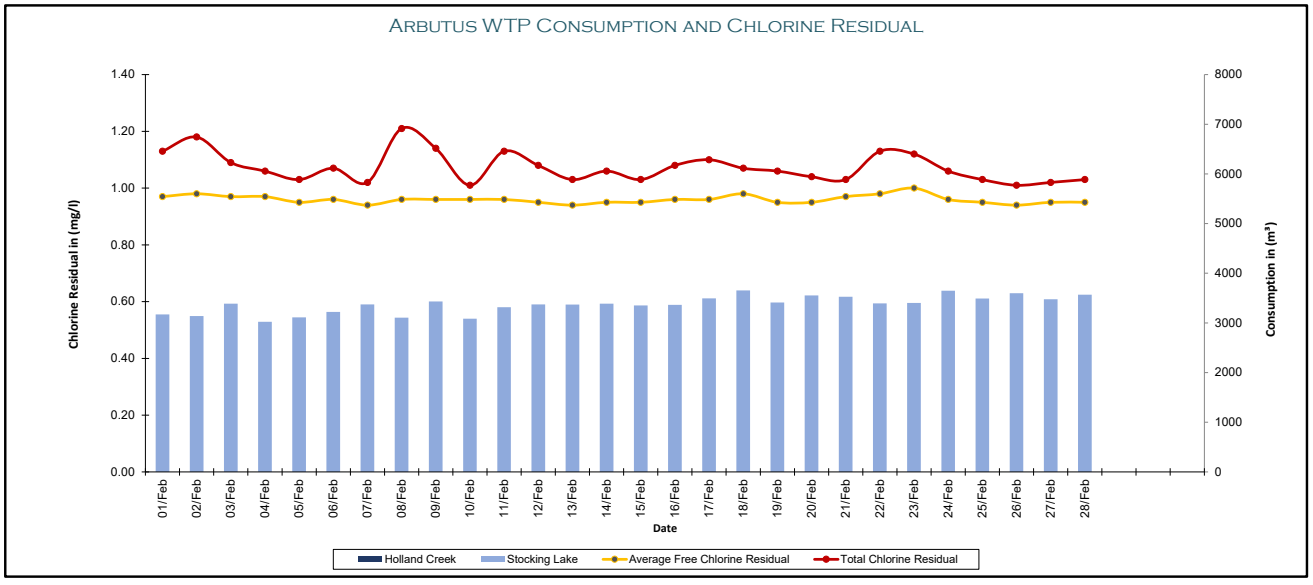
# Town of Ladysmith - Arbutus Water treatment Plant

## February 2025 - Monthly Report

| Date | Daily Flow | | | Chlorine Residual | | | | CT* | External Lab Testing | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Stocking Lake | Holland Creek | Combined Flow | Free Min | Free Max | Free Avg | Total | | HPC | E.coli | Total Coliforms | Aluminum | THM | HAA |
| | m³ | m³ | m³ | mg/l | mg/l | mg/l | mg/l | Minutes·mg/l | CFU | MPN | MPN | mg/l | mg/l | mg/l |
| 01-Feb | 3170 | 0 | 3170 | 1.03 | 1.13 | 0.97 | 1.13 | 148 | | | | | | |
| 02-Feb | 3139 | 0 | 3139 | 0.99 | 1.18 | 0.98 | 1.18 | 134 | | | | | | |
| 03-Feb | 3386 | 0 | 3386 | 0.96 | 1.09 | 0.97 | 1.09 | 127 | | | | | | |
| 04-Feb | 3022 | 0 | 3022 | 0.96 | 1.06 | 0.97 | 1.06 | 160 | | | | | | |
| 05-Feb | 3111 | 0 | 3111 | 0.94 | 1.03 | 0.95 | 1.03 | 161 | | | | | | |
| 06-Feb | 3220 | 0 | 3220 | 0.97 | 1.07 | 0.96 | 1.07 | 161 | | | | | | |
| 07-Feb | 3373 | 0 | 3373 | 1.00 | 1.02 | 0.94 | 1.02 | 114 | | | | | | |
| 08-Feb | 3105 | 0 | 3105 | 0.94 | 1.21 | 0.96 | 1.21 | 161 | < 1 | < 1 | < 1 | 0.0060 | 0.0165 | 0.0101 |
| 09-Feb | 3433 | 0 | 3433 | 0.98 | 1.14 | 0.96 | 1.14 | 154 | | | | | | |
| 10-Feb | 3084 | 0 | 3084 | 0.97 | 1.01 | 0.96 | 1.01 | 160 | | | | | | |
| 11-Feb | 3315 | 0 | 3315 | 0.98 | 1.13 | 0.96 | 1.13 | 153 | | | | | | |
| 12-Feb | 3374 | 0 | 3374 | 0.94 | 1.08 | 0.95 | 1.08 | 136 | | | | | | |
| 13-Feb | 3369 | 0 | 3369 | 0.93 | 1.03 | 0.94 | 1.03 | 144 | | | | | | |
| 14-Feb | 3386 | 0 | 3386 | 0.96 | 1.06 | 0.95 | 1.06 | 150 | < 1 | < 1 | < 1 | | | |
| 15-Feb | 3353 | 0 | 3353 | 0.95 | 1.03 | 0.95 | 1.03 | 160 | | | | | | |
| 16-Feb | 3364 | 0 | 3364 | 0.97 | 1.08 | 0.96 | 1.08 | 161 | | | | | | |
| 17-Feb | 3495 | 0 | 3495 | 0.96 | 1.01 | 0.96 | 1.10 | 150 | | | | | | |
| 18-Feb | 3654 | 0 | 3654 | 0.98 | 1.07 | 0.98 | 1.07 | 143 | | | | | | |
| 19-Feb | 3412 | 0 | 3412 | 0.94 | 1.06 | 0.95 | 1.06 | 128 | | | | | | |
| 20-Feb | 3552 | 0 | 3552 | 0.97 | 1.04 | 0.95 | 1.04 | 126 | | | | | | |
| 21-Feb | 3526 | 0 | 3526 | 0.99 | 1.03 | 0.97 | 1.03 | 135 | < 1 | < 1 | < 1 | | | |
| 22-Feb | 3393 | 0 | 3393 | 1.00 | 1.13 | 0.98 | 1.13 | 155 | | | | | | |
| 23-Feb | 3403 | 0 | 3403 | 1.01 | 1.12 | 1.00 | 1.12 | 123 | | | | | | |
| 24-Feb | 3647 | 0 | 3647 | 1.01 | 1.06 | 0.96 | 1.06 | 120 | | | | | | |
| 25-Feb | 3491 | 0 | 3491 | 0.96 | 1.03 | 0.95 | 1.03 | 125 | | | | | | |
| 26-Feb | 3599 | 0 | 3599 | 0.93 | 1.01 | 0.94 | 1.01 | 121 | | | | | | |
| 27-Feb | 3477 | 0 | 3477 | 0.95 | 1.02 | 0.95 | 1.02 | 140 | | | | | | |
| 28-Feb | 3569 | 0 | 3569 | 0.98 | 1.03 | 0.95 | 1.03 | 134 | < 1 | < 1 | < 1 | | | |
| | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | |
| **Total** | 94422 | 0 | 94422 | | | | | | | | | | | |
| **Average** | 3372 | 0 | 3372 | 0.97 | 1.07 | 0.96 | 1.07 | 142 | < 1 | < 1 | < 1 | 0.0060 | 0.0165 | 0.01010 |

*CT - Recorded as the minimum value at the highest daily flow ** Manual Residual were not taken



VEOLIA

InSight

# Town of Ladysmith Arbutus DWTP

## Monthly LRV and Turbidity Report

# 01/29/2025 - 03/01/2025

## LRV Monthly Average

| Asset | Parameter | Health | Avg | Std. Dev | Points | LL | L | %In | % between L and LL | % < LL | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|
| UF 1 | LRV | | 5.0 | 0.0 | 32 | 4.25 | 4.5 | 100 % | 0 % | 0 % | # |
| UF 2 | LRV | | 5.0 | 0.0 | 33 | 4.25 | 4.5 | 100 % | 0 % | 0 % | # |
| UF 3 | LRV | | 5.0 | 0.0 | 32 | 4.25 | 4.5 | 100 % | 0 % | 0 % | # |

## LRV Daily Values

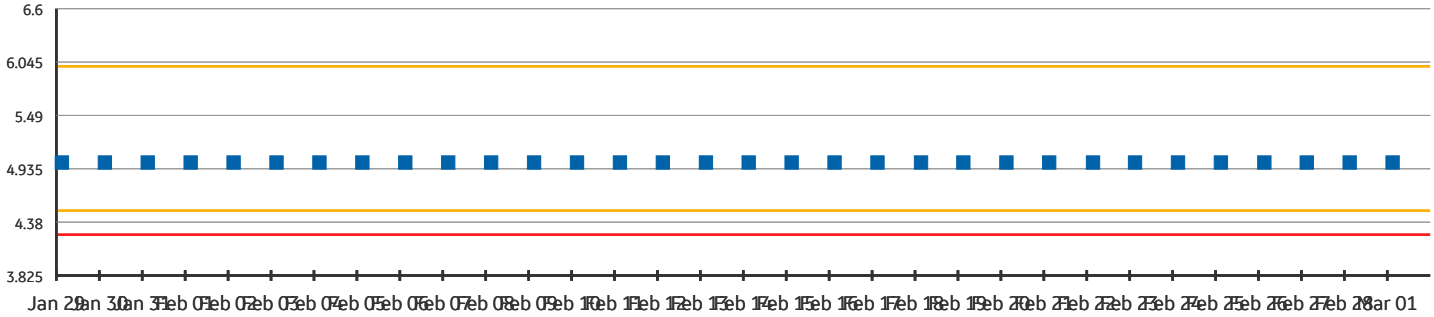
| Asset | Parameter | Jan 29 | Jan 30 | Jan 31 | Feb 01 | Feb 02 | Feb 03 | Feb 04 | Feb 05 | Feb 06 | Feb 07 | Feb 08 | Feb 09 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UF 1 | LRV | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 |
| UF 2 | LRV | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 |
| UF 3 | LRV | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 |

| Asset | Feb 10 | Feb 11 | Feb 12 | Feb 13 | Feb 14 | Feb 15 | Feb 16 | Feb 17 | Feb 18 | Feb 19 | Feb 20 | Feb 21 | Feb 22 | Feb 23 | Feb 24 | Feb 25 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UF 1 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 |
| UF 2 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 |
| UF 3 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 |

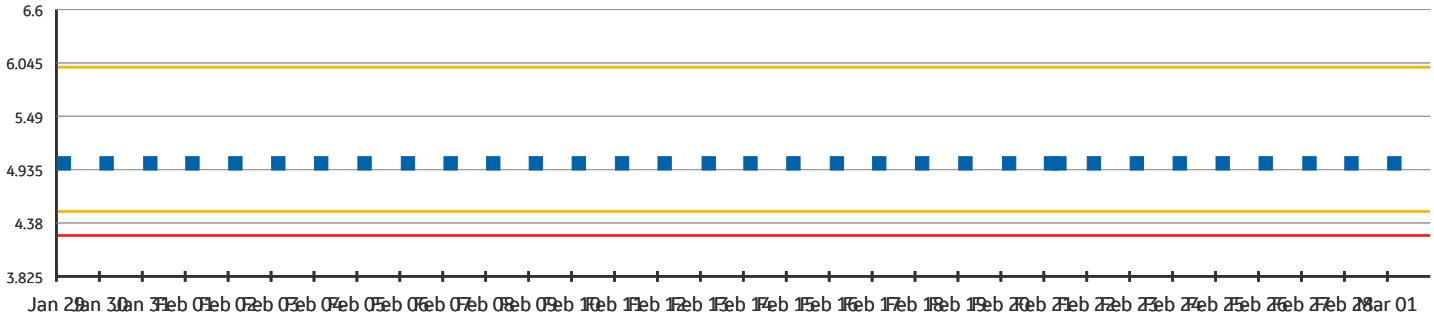
| Asset | Feb 26 | Feb 27 | Feb 28 | Mar 01 |
|---|---|---|---|---|
| UF 1 | 5.0 | 5.0 | 5.0 | 5.0 |
| UF 2 | 5.0 | 5.0 | 5.0 | 5.0 |
| UF 3 | 5.0 | 5.0 | 5.0 | 5.0 |

## LRV Raw Data

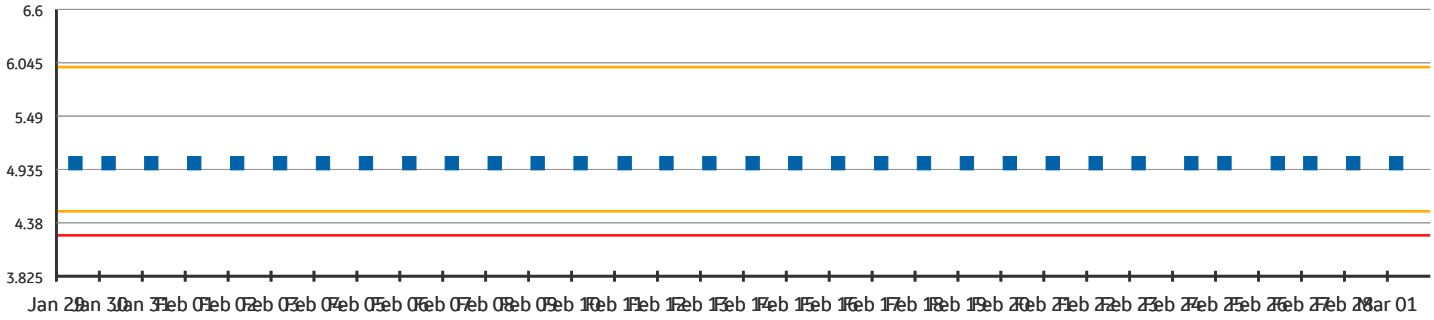
## UF 1 - LRV (#)




## UF 2 - LRV (#)




## UF 3 - LRV (#)




## Turbidity Monthly Average

| Asset | Parameter | Health | Avg | Std. Dev | Points | H | HH | %In | % between H and HH | % > HH | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|
| UF 1 | PermeateTurbidity | | 0.015 | 0.0 | 45214 | -- | -- | 100 % | 0 % | 0 % | NTU |
| UF 2 | PermeateTurbidity | | 0.015 | 0.0 | 45214 | -- | -- | 100 % | 0 % | 0 % | NTU |
| UF 3 | PermeateTurbidity | | 0.018 | 0.0 | 45214 | -- | -- | 100 % | 0 % | 0 % | NTU |
| UF 1 | PermeateTurbidityAfterBP | ● | 0.015 | 0.0 | 386 | 0.1 | 0.3 | 100 % | 0 % | 0 % | NTU |
| UF 2 | PermeateTurbidityAfterBP | ● | 0.015 | 0.0 | 393 | 0.1 | 0.3 | 100 % | 0 % | 0 % | NTU |
| UF 3 | PermeateTurbidityAfterBP | ● | 0.018 | 0.0 | 375 | 0.1 | 0.3 | 100 % | 0 % | 0 % | NTU |

## Turbidity Daily Averages

| Asset | Parameter | Jan 29 | Jan 30 | Jan 31 | Feb 01 | Feb 02 | Feb 03 | Feb 04 | Feb 05 | Feb 06 | Feb 07 | Feb 08 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UF 1 | PermeateTurbidity | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 2 | PermeateTurbidity | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 3 | PermeateTurbidity | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 |

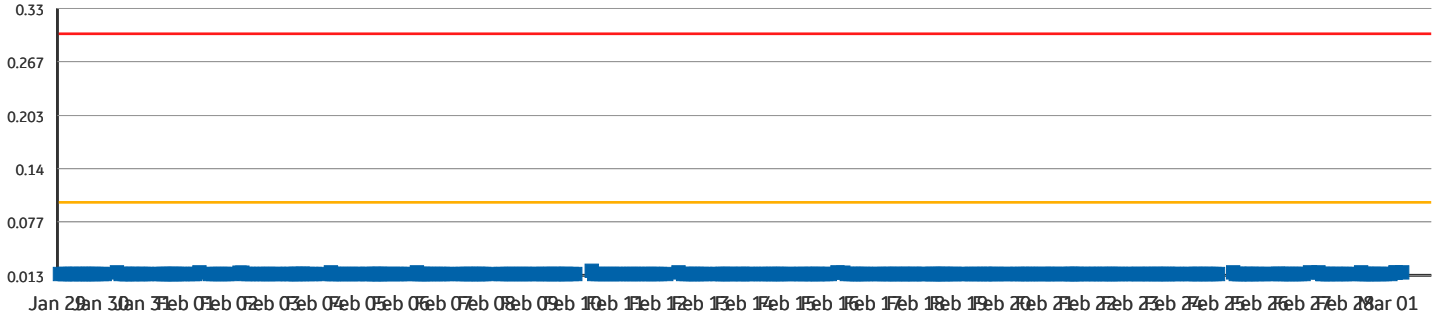
| Asset | Parameter | Jan 29 | Jan 30 | Jan 31 | Feb 01 | Feb 02 | Feb 03 | Feb 04 | Feb 05 | Feb 06 | Feb 07 | Feb 08 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UF 1 | PermeateTurbidityAfterBP | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 2 | PermeateTurbidityAfterBP | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 3 | PermeateTurbidityAfterBP | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 | 0.017 |

| Asset | Feb 09 | Feb 10 | Feb 11 | Feb 12 | Feb 13 | Feb 14 | Feb 15 | Feb 16 | Feb 17 | Feb 18 | Feb 19 | Feb 20 | Feb 21 | Feb 22 | Feb 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UF 1 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 2 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 3 | 0.017 | 0.017 | 0.017 | 0.017 | 0.018 | 0.018 | 0.019 | 0.018 | 0.018 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 |
| UF 1 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 2 | 0.016 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 3 | 0.017 | 0.018 | 0.018 | 0.018 | 0.018 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 |

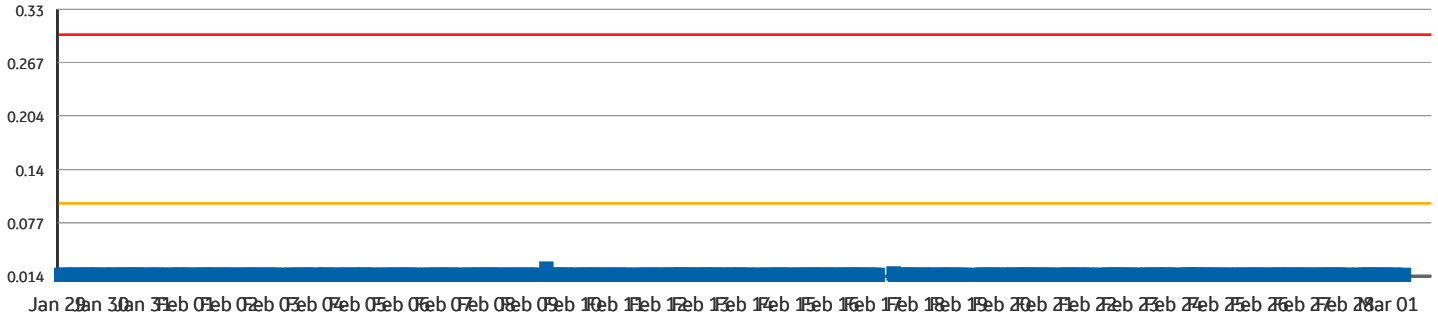
| Asset | Feb 24 | Feb 25 | Feb 26 | Feb 27 | Feb 28 | Mar 01 |
|---|---|---|---|---|---|---|
| UF 1 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 2 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 3 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 |
| UF 1 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.016 |
| UF 2 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 | 0.015 |
| UF 3 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 | 0.019 |

## Turbidity Raw Data

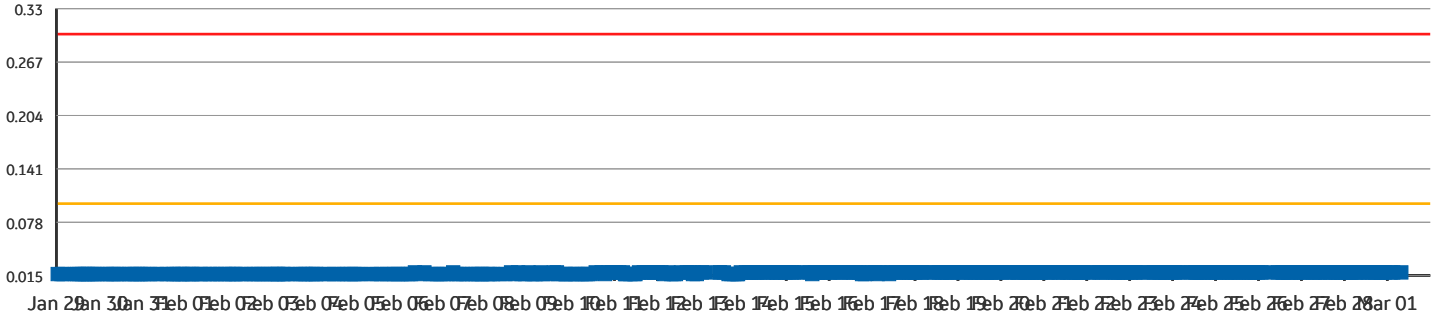
### UF 1 - PermeateTurbidityAfterBP (NTU)




### UF 2 - PermeateTurbidityAfterBP (NTU)



# UF 3 - PermeateTurbidityAfterBP (NTU)



# CERTIFICATE OF ANALYSIS

| | | | | | |
|---|---|---|---|---|---|
| **Work Order** | : | **VA25A2735** | | | |
| Client | : | **Town of Ladysmith** | Laboratory | : | ALS Environmental - Vancouver |
| Contact | : | Shawn Baker | Account Manager | : | Kevin Bhikadia |
| Address | : | 410 Esplanade PO Box 220<br>Ladysmith British Columbia Canada V9G 1A2 | Address | : | 8081 Lougheed Highway<br>Burnaby BC Canada V5A 1W9 |
| Telephone | : | ---- | Telephone | : | +1 604 253 4188 |
| Project | : | Arbutus Water Treatment Plant - Weekly Sampling | Date Samples Received | : | 07-Feb-2025 13:30 |
| PO | : | PO #10940 | Date Analysis Commenced | : | 07-Feb-2025 |
| C-O-C number | : | ---- | Issue Date | : | 13-Feb-2025 16:08 |
| Sampler | : | ---- | | | |
| Site | : | Town of Ladysmith | | | |
| Quote number | : | Town of Ladysmith Standing Offer | | | |
| No. of samples received | : | 4 | | | |
| No. of samples analysed | : | 4 | | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

## *Signatories*

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Lindsay Gung | Supervisor - Water Chemistry | Microbiology, Burnaby, British Columbia |
| Lindsay Gung | Supervisor - Water Chemistry | Inorganics, Burnaby, British Columbia |
| Monica Ko | Lab Assistant | Inorganics, Burnaby, British Columbia |

## General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key: CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances.
LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| CFU/mL | colony forming units per millilitre |
| CU | colour units (1 cu = 1 mg/l pt) |
| mg/L | milligrams per litre |
| MPN/100mL | most probable number per hundred millilitres |
| NTU | nephelometric turbidity units |
| pH units | pH units |
| µS/cm | microsiemens per centimetre |

<: less than.
>: greater than.
Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.
UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

## Workorder Comments

**Sample Treated Water (post reservoir): HPC Exceeded Recommended Holding Time prior to receipt at the lab for Microbiology analysis. Testing will proceed.**

## Qualifiers

| *Qualifier* | Description |
|---|---|
| SFP | Sample was filtered and preserved at the laboratory. |

## Analytical Results

**Sub-Matrix: Water**
**(Matrix: Water)**

| Analyte | CAS Number | Method/Lab | LOR | Unit | Client sample ID: Raw Water | DAF Effluent | UF Effluent | Treated Water (post reservoir) | ---- |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Client sampling date / time: 06-Feb-2025 10:30 | 06-Feb-2025 10:30 | 06-Feb-2025 10:30 | 06-Feb-2025 10:30 | ---- |
| | | | | | VA25A2735-001 | VA25A2735-002 | VA25A2735-003 | VA25A2735-004 | ---- |
| | | | | | Result | Result | Result | Result | ---- |
| **Physical Tests** | | | | | | | | | |
| **Alkalinity, total (as CaCO3)** | ---- | E290/VA | 1.0 | mg/L | ---- | ---- | ---- | 15.8 | ---- |
| **Colour, true** | ---- | E329/VA | 5.0 | CU | ---- | ---- | ---- | <5.0 | ---- |
| **Conductivity** | ---- | E100/VA | 2.0 | µS/cm | ---- | ---- | ---- | 63.9 | ---- |
| **pH** | ---- | E108/VA | 0.10 | pH units | ---- | ---- | ---- | 7.08 | ---- |
| **Turbidity** | ---- | E121/VA | 0.10 | NTU | ---- | ---- | ---- | <0.10 | ---- |
| **Organic / Inorganic Carbon** | | | | | | | | | |
| **Carbon, dissolved organic [DOC]** | ---- | E358-L/VA | 0.50 | mg/L | 1.24 SFP | 1.42 | 1.35 SFP | ---- | ---- |
| **Carbon, total organic [TOC]** | ---- | E355-L/VA | 0.50 | mg/L | 2.66 | 1.20 | 1.15 | ---- | ---- |
| **Microbiological Tests** | | | | | | | | | |
| **Heterotrophic plate count [HPC]** | ---- | E020/VA | 1 | CFU/mL | ---- | ---- | ---- | <1 | ---- |
| **Coliforms, Escherichia coli [E. coli]** | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |
| **Coliforms, total** | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |

Please refer to the General Comments section for an explanation of any result qualifiers detected.

ALS Canada Ltd.

right solutions.
right partner.

# QUALITY CONTROL INTERPRETIVE REPORT

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A2735** | Page | : 1 of 7 |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith BC Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby, British Columbia Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment Plant - Weekly Sampling | Date Samples Received | : 07-Feb-2025 13:30 |
| PO | : PO #10940 | Issue Date | : 13-Feb-2025 16:08 |
| C-O-C number | : ---- | | |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer_V2 | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

**This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.**

**Key**

**Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.**
**CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.**
**DQO: Data Quality Objective.**
**LOR: Limit of Reporting (detection limit).**
**RPD: Relative Percent Difference.**

## *Workorder Comments*

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

## *Summary of Outliers*

### *Outliers : Quality Control Samples*

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

### *Outliers: Reference Material (RM) Samples*

- No Reference Material (RM) Sample outliers occur.

## *Outliers : Analysis Holding Time Compliance (Breaches)*

- Analysis Holding Time Outliers exist - please see following pages for full details.

## *Outliers : Frequency of Quality Control Samples*

- Quality Control Sample Frequency Outliers occur - please see following pages for full details.

# CERTIFICATE OF ANALYSIS

| | | | | |
|---|---|---|---|---|
| **Work Order** | : | **VA25A3032** | | |
| Client | : | **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : | Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : | 410 Esplanade PO Box 220<br>Ladysmith British Columbia Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby BC Canada V5A 1W9 |
| Telephone | : | ---- | Telephone | : +1 604 253 4188 |
| Project | : | Arbutus Water Treatment Plant - Monthly Sampling | Date Samples Received | : 12-Feb-2025 12:20 |
| PO | : | PO #10940 | Date Analysis Commenced | : 12-Feb-2025 |
| C-O-C number | : | ---- | Issue Date | : 26-Feb-2025 16:22 |
| Sampler | : | ---- | | |
| Site | : | Town of Ladysmith | | |
| Quote number | : | Town of Ladysmith Standing Offer | | |
| No. of samples received | : | 1 | | |
| No. of samples analysed | : | 1 | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

## *Signatories*

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Kim Jensen | Department Manager - Metals | Metals, Burnaby, British Columbia |
| Ophelia Chiu | Department Manager - Organics | Organics, Burnaby, British Columbia |
| Sanja Risticevic | Department Manager - LCMS | LCMS, Waterloo, Ontario |

## General Comments

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Key: CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances.
LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| mg/L | milligrams per litre |
| µg/L | micrograms per litre |

<: less than.
>: greater than.
Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.
UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

## Analytical Results

**Sub-Matrix: Water**
**(Matrix: Water)**

| Analyte | CAS Number | Method/Lab | LOR | Unit | Client sample ID | Treated Water (post reservoir) | ---- | ---- | ---- | ---- |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Client sampling date / time | 11-Feb-2025 10:30 | ---- | ---- | ---- | ---- |
| | | | | | | VA25A3032-001 | ---- | ---- | ---- | ---- |
| | | | | | | Result | ---- | ---- | ---- | ---- |
| **Total Metals** | | | | | | | | | | |
| **Aluminum, total** | 7429-90-5 | E420/VA | 0.0030 | mg/L | | 0.0060 | ---- | ---- | ---- | ---- |
| **Volatile Organic Compounds [THMs]** | | | | | | | | | | |
| **Bromodichloromethane** | 75-27-4 | E611B/VA | 1.0 | µg/L | | 1.1 | ---- | ---- | ---- | ---- |
| **Bromoform** | 75-25-2 | E611B/VA | 1.0 | µg/L | | <1.0 | ---- | ---- | ---- | ---- |
| **Chloroform** | 67-66-3 | E611B/VA | 1.0 | µg/L | | 15.4 | ---- | ---- | ---- | ---- |
| **Dibromochloromethane** | 124-48-1 | E611B/VA | 1.0 | µg/L | | <1.0 | ---- | ---- | ---- | ---- |
| **Trihalomethanes [THMs], total** | ---- | E611B/VA | 2.0 | µg/L | | 16.5 | ---- | ---- | ---- | ---- |
| **Volatile Organic Compounds [THMs] Surrogates** | | | | | | | | | | |
| **Bromofluorobenzene, 4-** | 460-00-4 | E611B/VA | 1.0 | % | | 101 | ---- | ---- | ---- | ---- |
| **Difluorobenzene, 1,4-** | 540-36-3 | E611B/VA | 1.0 | % | | 102 | ---- | ---- | ---- | ---- |
| **Haloacetic Acids** | | | | | | | | | | |
| **Bromochloroacetic acid** | 5589-96-8 | E750/WT | 1.00 | µg/L | | <1.00 | ---- | ---- | ---- | ---- |
| **Dibromoacetic acid** | 631-64-1 | E750/WT | 1.00 | µg/L | | <1.00 | ---- | ---- | ---- | ---- |
| **Dichloroacetic acid** | 79-43-6 | E750/WT | 1.00 | µg/L | | 4.58 | ---- | ---- | ---- | ---- |
| **Monobromoacetic acid** | 79-08-3 | E750/WT | 1.00 | µg/L | | <1.00 | ---- | ---- | ---- | ---- |
| **Monochloroacetic acid** | 79-11-8 | E750/WT | 1.00 | µg/L | | <1.00 | ---- | ---- | ---- | ---- |
| **Trichloroacetic acid** | 76-03-9 | E750/WT | 1.00 | µg/L | | 5.54 | ---- | ---- | ---- | ---- |
| **Haloacetic acids, total [HAA5]** | n/a | E750/WT | 5.00 | µg/L | | 10.1 | ---- | ---- | ---- | ---- |

Please refer to the General Comments section for an explanation of any result qualifiers detected.

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# QUALITY CONTROL INTERPRETIVE REPORT

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A3032** | Page | : 1 of 5 |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith BC Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby, British Columbia Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment Plant - Monthly Sampling | Date Samples Received | : 12-Feb-2025 12:20 |
| PO | : PO #10940 | Issue Date | : 26-Feb-2025 16:22 |
| C-O-C number | : ---- | | |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer_V2 | | |
| No. of samples received | : 1 | | |
| No. of samples analysed | : 1 | | |

**This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.**

**Key**

**Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.**
**CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.**
**DQO: Data Quality Objective.**
**LOR: Limit of Reporting (detection limit).**
**RPD: Relative Percent Difference.**

## *Workorder Comments*

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

## *Summary of Outliers*

### *Outliers : Quality Control Samples*

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Matrix Spike Duplicate (MSD) outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

### *Outliers: Reference Material (RM) Samples*

- No Reference Material (RM) Sample outliers occur.

## *Outliers : Analysis Holding Time Compliance (Breaches)*

- No Analysis Holding Time Outliers exist.

## *Outliers : Frequency of Quality Control Samples*

- No Quality Control Sample Frequency Outliers occur.

# CERTIFICATE OF ANALYSIS

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A3026** | | |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith British Columbia Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby BC Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment - Weekly Sampling | Date Samples Received | : 12-Feb-2025 12:30 |
| PO | : 10880 | Date Analysis Commenced | : 12-Feb-2025 |
| C-O-C number | : ---- | Issue Date | : 20-Feb-2025 15:27 |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

## *Signatories*

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Caitlin Macey | Team Leader - Inorganics | Microbiology, Burnaby, British Columbia |
| Lindsay Gung | Supervisor - Water Chemistry | Inorganics, Burnaby, British Columbia |
| Robin Weeks | Team Leader - Metals | Microbiology, Burnaby, British Columbia |

## General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key: CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances.
LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| CFU/mL | colony forming units per millilitre |
| CU | colour units (1 cu = 1 mg/l pt) |
| mg/L | milligrams per litre |
| MPN/100mL | most probable number per hundred millilitres |
| NTU | nephelometric turbidity units |
| pH units | pH units |
| µS/cm | microsiemens per centimetre |

<: less than.
>: greater than.
Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.
UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

## Workorder Comments

**HPC Exceeded Recommended Holding Time prior to receipt at the lab for Microbiology analysis.**

## Analytical Results

| Sub-Matrix: Water (Matrix: Water) | | | | Client sample ID | Raw Water | DAF Effluent | UF Effluent | Treated Water (post reservoir) | ---- |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Client sampling date / time | 11-Feb-2025 10:30 | 11-Feb-2025 10:30 | 11-Feb-2025 10:30 | 11-Feb-2025 10:30 | ---- |
| Analyte | CAS Number | Method/Lab | LOR | Unit | VA25A3026-001 | VA25A3026-002 | VA25A3026-003 | VA25A3026-004 | ---- |
| | | | | | Result | Result | Result | Result | ---- |
| **Physical Tests** | | | | | | | | | |
| Alkalinity, total (as CaCO3) | ---- | E290/VA | 1.0 | mg/L | ---- | ---- | ---- | 17.2 | ---- |
| Colour, true | ---- | E329/VA | 5.0 | CU | ---- | ---- | ---- | <5.0 | ---- |
| Conductivity | ---- | E100/VA | 2.0 | µS/cm | ---- | ---- | ---- | 65.8 | ---- |
| pH | ---- | E108/VA | 0.10 | pH units | ---- | ---- | ---- | 7.44 | ---- |
| Turbidity | ---- | E121/VA | 0.10 | NTU | ---- | ---- | ---- | <0.10 | ---- |
| **Organic / Inorganic Carbon** | | | | | | | | | |
| Carbon, dissolved organic [DOC] | ---- | E358-L/VA | 0.50 | mg/L | 3.70 | 1.63 | 1.57 | ---- | ---- |
| Carbon, total organic [TOC] | ---- | E355-L/VA | 0.50 | mg/L | 3.11 | 1.41 | 1.17 | ---- | ---- |
| **Microbiological Tests** | | | | | | | | | |
| Heterotrophic plate count [HPC] | ---- | E020/VA | 1 | CFU/mL | ---- | ---- | ---- | <1 | ---- |
| Coliforms, Escherichia coli [E. coli] | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |
| Coliforms, total | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |

Please refer to the General Comments section for an explanation of any result qualifiers detected.

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# QUALITY CONTROL INTERPRETIVE REPORT

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A3026** | Page | : 1 of 7 |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith BC Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby, British Columbia Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment - Weekly Sampling | Date Samples Received | : 12-Feb-2025 12:30 |
| PO | : 10880 | Issue Date | : 20-Feb-2025 15:28 |
| C-O-C number | : ---- | | |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer_V2 | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

**This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.**

**Key**

**Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.**
**CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.**
**DQO: Data Quality Objective.**
**LOR: Limit of Reporting (detection limit).**
**RPD: Relative Percent Difference.**

## *Workorder Comments*

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

## *Summary of Outliers*

### *Outliers : Quality Control Samples*

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

### *Outliers: Reference Material (RM) Samples*

- No Reference Material (RM) Sample outliers occur.

## Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

## Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

# CERTIFICATE OF ANALYSIS

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A3443** | | |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith British Columbia Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby BC Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment Plant - Weekly Sampling | Date Samples Received | : 19-Feb-2025 09:50 |
| PO | : PO #10940 | Date Analysis Commenced | : 19-Feb-2025 |
| C-O-C number | : ---- | Issue Date | : 25-Feb-2025 09:34 |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

## *Signatories*

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Lindsay Gung | Supervisor - Water Chemistry | Inorganics, Burnaby, British Columbia |
| Lindsay Gung | Supervisor - Water Chemistry | Microbiology, Burnaby, British Columbia |

## General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key: CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances.
LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| CFU/mL | colony forming units per millilitre |
| CU | colour units (1 cu = 1 mg/l pt) |
| mg/L | milligrams per litre |
| MPN/100mL | most probable number per hundred millilitres |
| NTU | nephelometric turbidity units |
| pH units | pH units |
| µS/cm | microsiemens per centimetre |

<: less than.
>: greater than.
Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.
UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

## Analytical Results

| Sub-Matrix: Water (Matrix: Water) | | | | Client sample ID | Raw Water | DAF Effluent | UF Effluent | Treated Water (post reservoir) | ---- |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Client sampling date / time | 18-Feb-2025 10:30 | 18-Feb-2025 10:30 | 18-Feb-2025 10:30 | 18-Feb-2025 10:30 | ---- |
| Analyte | CAS Number | Method/Lab | LOR | Unit | VA25A3443-001 | VA25A3443-002 | VA25A3443-003 | VA25A3443-004 | ---- |
| | | | | | Result | Result | Result | Result | ---- |
| **Physical Tests** | | | | | | | | | |
| Alkalinity, total (as CaCO3) | ---- | E290/VA | 1.0 | mg/L | ---- | ---- | ---- | 16.1 | ---- |
| Colour, true | ---- | E329/VA | 5.0 | CU | ---- | ---- | ---- | <5.0 | ---- |
| Conductivity | ---- | E100/VA | 2.0 | µS/cm | ---- | ---- | ---- | 68.5 | ---- |
| pH | ---- | E108/VA | 0.10 | pH units | ---- | ---- | ---- | 7.49 | ---- |
| Turbidity | ---- | E121/VA | 0.10 | NTU | ---- | ---- | ---- | 0.12 | ---- |
| **Organic / Inorganic Carbon** | | | | | | | | | |
| Carbon, dissolved organic [DOC] | ---- | E358-L/VA | 0.50 | mg/L | 2.95 | 1.28 | 1.34 | ---- | ---- |
| Carbon, total organic [TOC] | ---- | E355-L/VA | 0.50 | mg/L | 3.51 | 1.46 | 1.22 | ---- | ---- |
| **Microbiological Tests** | | | | | | | | | |
| Heterotrophic plate count [HPC] | ---- | E020/VA | 1 | CFU/mL | ---- | ---- | ---- | <1 | ---- |
| Coliforms, Escherichia coli [E. coli] | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |
| Coliforms, total | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |

Please refer to the General Comments section for an explanation of any result qualifiers detected.

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# QUALITY CONTROL INTERPRETIVE REPORT

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A3443** | Page | : 1 of 7 |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith BC Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby, British Columbia Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment Plant - Weekly Sampling | Date Samples Received | : 19-Feb-2025 09:50 |
| PO | : PO #10940 | Issue Date | : 25-Feb-2025 09:34 |
| C-O-C number | : ---- | | |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer_V2 | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

**This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.**

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**CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.**
**DQO: Data Quality Objective.**
**LOR: Limit of Reporting (detection limit).**
**RPD: Relative Percent Difference.**

## *Workorder Comments*

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

## *Summary of Outliers*

### *Outliers : Quality Control Samples*

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

### *Outliers: Reference Material (RM) Samples*

- No Reference Material (RM) Sample outliers occur.

## Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

## Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

# CERTIFICATE OF ANALYSIS

| | | | | |
|---|---|---|---|---|
| **Work Order** | : | **VA25A3712** | | |
| Client | : | **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : | Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : | 410 Esplanade PO Box 220<br>Ladysmith British Columbia Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby BC Canada V5A 1W9 |
| Telephone | : | ---- | Telephone | : +1 604 253 4188 |
| Project | : | Arbutus Water Treatment Plant - Quarterly Lake Sampling | Date Samples Received | : 21-Feb-2025 11:02 |
| PO | : | 10880 | Date Analysis Commenced | : 21-Feb-2025 |
| C-O-C number | : | ---- | Issue Date | : 04-Mar-2025 19:47 |
| Sampler | : | ---- | | |
| Site | : | RFP Tender No. 2022-IS-20 Extension | | |
| Quote number | : | Town of Ladysmith Standing Offer | | |
| No. of samples received | : | 2 | | |
| No. of samples analysed | : | 2 | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

## *Signatories*

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Asha Tauckoor | Laboratory Analyst | Organics, Burnaby, British Columbia |
| Caitlin Macey | Team Leader - Inorganics | Microbiology, Burnaby, British Columbia |
| Kevin Duarte | Supervisor - Metals ICP Instrumentation | Metals, Burnaby, British Columbia |
| Lindsay Gung | Supervisor - Water Chemistry | Inorganics, Burnaby, British Columbia |

## General Comments

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Key: CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances.
LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| % T/cm | % transmittance per centimetre |
| AU/cm | absorbance units per centimetre |
| CU | colour units (1 cu = 1 mg/l pt) |
| mg/L | milligrams per litre |
| MPN/100mL | most probable number per hundred millilitres |
| NTU | nephelometric turbidity units |
| pH units | pH units |
| µg/L | micrograms per litre |

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>: greater than.
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# CERTIFICATE OF ANALYSIS

| | | | | | |
|---|---|---|---|---|---|
| **Work Order** | : | **VA25A4115** | | | |
| Client | : | **Town of Ladysmith** | Laboratory | : | ALS Environmental - Vancouver |
| Contact | : | Shawn Baker | Account Manager | : | Kevin Bhikadia |
| Address | : | 410 Esplanade PO Box 220<br>Ladysmith British Columbia Canada V9G 1A2 | Address | : | 8081 Lougheed Highway<br>Burnaby BC Canada V5A 1W9 |
| Telephone | : | ---- | Telephone | : | +1 604 253 4188 |
| Project | : | Arbutus Water Treatment Plant - Quarterly DT Sampling | Date Samples Received | : | 26-Feb-2025 11:25 |
| PO | : | 10880 | Date Analysis Commenced | : | 27-Feb-2025 |
| C-O-C number | : | ---- | Issue Date | : | 05-Mar-2025 15:38 |
| Sampler | : | ---- | | | |
| Site | : | RFP Tender No. 2022-IS-20 Extension | | | |
| Quote number | : | Town of Ladysmith Standing Offer | | | |
| No. of samples received | : | 4 | | | |
| No. of samples analysed | : | 4 | | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

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- Analytical Results

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| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Kim Jensen | Department Manager - Metals | Metals, Burnaby, British Columbia |
| Lindsay Gung | Supervisor - Water Chemistry | Inorganics, Burnaby, British Columbia |

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Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key: CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances.
LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| mg/L | milligrams per litre |
| pH units | pH units |

<: less than.
>: greater than.
Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.
UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

## Analytical Results

**Sub-Matrix: Water**
**(Matrix: Water)**

| Analyte | CAS Number | Method/Lab | LOR | Unit | FJCC | Town Hall | Fire Depatment | RCMP | ---- |
|---|---|---|---|---|---|---|---|---|---|
| *Client sample ID* | | | | | | | | | |
| *Client sampling date / time* | | | | | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | ---- |
| | | | | | VA25A4115-001 | VA25A4115-002 | VA25A4115-003 | VA25A4115-004 | ---- |
| | | | | | Result | Result | Result | Result | ---- |
| **Physical Tests** | | | | | | | | | |
| **Alkalinity, total (as CaCO3)** | ---- | E290/VA | 1.0 | mg/L | 17.4 | 16.7 | 17.3 | 17.5 | ---- |
| **pH** | ---- | E108/VA | 0.10 | pH units | 7.46 | 7.35 | 7.43 | 7.44 | ---- |
| **Solids, total dissolved [TDS]** | ---- | E162/VA | 10 | mg/L | 44 | 47 | 44 | 46 | ---- |
| **Total Metals** | | | | | | | | | |
| **Aluminum, total** | 7429-90-5 | E420/VA | 0.0030 | mg/L | 0.0052 | 0.0061 | 0.0065 | 0.0149 | ---- |
| **Antimony, total** | 7440-36-0 | E420/VA | 0.00010 | mg/L | <0.00010 | <0.00010 | <0.00010 | <0.00010 | ---- |
| **Arsenic, total** | 7440-38-2 | E420/VA | 0.00010 | mg/L | <0.00010 | <0.00010 | <0.00010 | <0.00010 | ---- |
| **Barium, total** | 7440-39-3 | E420/VA | 0.00010 | mg/L | 0.00284 | 0.00288 | 0.00320 | 0.00308 | ---- |
| **Beryllium, total** | 7440-41-7 | E420/VA | 0.000020 | mg/L | <0.000020 | <0.000020 | <0.000020 | <0.000020 | ---- |
| **Bismuth, total** | 7440-69-9 | E420/VA | 0.000050 | mg/L | <0.000050 | <0.000050 | <0.000050 | <0.000050 | ---- |
| **Boron, total** | 7440-42-8 | E420/VA | 0.010 | mg/L | 0.021 | 0.021 | 0.022 | 0.023 | ---- |
| **Cadmium, total** | 7440-43-9 | E420/VA | 0.0000050 | mg/L | <0.0000050 | <0.0000050 | <0.0000050 | <0.0000050 | ---- |
| **Calcium, total** | 7440-70-2 | E420/VA | 0.050 | mg/L | 3.55 | 3.53 | 3.70 | 3.42 | ---- |
| **Cesium, total** | 7440-46-2 | E420/VA | 0.000010 | mg/L | <0.000010 | <0.000010 | <0.000010 | <0.000010 | ---- |
| **Chromium, total** | 7440-47-3 | E420/VA | 0.00050 | mg/L | <0.00050 | <0.00050 | <0.00050 | <0.00050 | ---- |
| **Cobalt, total** | 7440-48-4 | E420/VA | 0.00010 | mg/L | <0.00010 | <0.00010 | <0.00010 | <0.00010 | ---- |
| **Copper, total** | 7440-50-8 | E420/VA | 0.00050 | mg/L | 0.0382 | 0.0526 | 0.203 | 0.478 | ---- |
| **Iron, total** | 7439-89-6 | E420/VA | 0.010 | mg/L | <0.010 | <0.010 | 0.012 | 0.210 | ---- |
| **Lead, total** | 7439-92-1 | E420/VA | 0.000050 | mg/L | 0.000510 | 0.000206 | 0.00145 | 0.00184 | ---- |
| **Lithium, total** | 7439-93-2 | E420/VA | 0.0010 | mg/L | <0.0010 | <0.0010 | <0.0010 | <0.0010 | ---- |
| **Magnesium, total** | 7439-95-4 | E420/VA | 0.0050 | mg/L | 0.412 | 0.421 | 0.415 | 0.552 | ---- |

## Analytical Results

**Sub-Matrix: Water**
**(Matrix: Water)**

| Analyte | CAS Number | Method/Lab | LOR | Unit | FJCC | Town Hall | Fire Depatment | RCMP | ---- |
|---|---|---|---|---|---|---|---|---|---|
| *Client sample ID* | | | | | | | | | |
| *Client sampling date / time* | | | | | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | ---- |
| | | | | | VA25A4115-001 | VA25A4115-002 | VA25A4115-003 | VA25A4115-004 | ---- |
| | | | | | Result | Result | Result | Result | ---- |
| **Total Metals** | | | | | | | | | |
| **Manganese, total** | 7439-96-5 | E420/VA | 0.00010 | mg/L | 0.00024 | 0.00030 | 0.00036 | 0.00167 | ---- |
| **Molybdenum, total** | 7439-98-7 | E420/VA | 0.000050 | mg/L | 0.000120 | 0.000128 | 0.000118 | 0.000116 | ---- |
| **Nickel, total** | 7440-02-0 | E420/VA | 0.00050 | mg/L | <0.00050 | <0.00050 | <0.00050 | 0.00076 | ---- |
| **Phosphorus, total** | 7723-14-0 | E420/VA | 0.050 | mg/L | <0.050 | <0.050 | <0.050 | <0.050 | ---- |
| **Potassium, total** | 7440-09-7 | E420/VA | 0.050 | mg/L | 0.223 | 0.220 | 0.221 | 0.230 | ---- |
| **Rubidium, total** | 7440-17-7 | E420/VA | 0.00020 | mg/L | 0.00050 | 0.00053 | 0.00052 | 0.00053 | ---- |
| **Selenium, total** | 7782-49-2 | E420/VA | 0.000050 | mg/L | <0.000050 | <0.000050 | <0.000050 | <0.000050 | ---- |
| **Silicon, total** | 7440-21-3 | E420/VA | 0.10 | mg/L | 1.79 | 1.76 | 1.81 | 1.75 | ---- |
| **Silver, total** | 7440-22-4 | E420/VA | 0.000010 | mg/L | <0.000010 | <0.000010 | <0.000010 | 0.000015 | ---- |
| **Sodium, total** | 7440-23-5 | E420/VA | 0.050 | mg/L | 8.73 | 8.62 | 8.59 | 8.57 | ---- |
| **Strontium, total** | 7440-24-6 | E420/VA | 0.00020 | mg/L | 0.0105 | 0.0104 | 0.0114 | 0.0106 | ---- |
| **Sulfur, total** | 7704-34-9 | E420/VA | 0.50 | mg/L | 0.50 | <0.50 | <0.50 | <0.50 | ---- |
| **Tellurium, total** | 13494-80-9 | E420/VA | 0.00020 | mg/L | <0.00020 | <0.00020 | <0.00020 | <0.00020 | ---- |
| **Thallium, total** | 7440-28-0 | E420/VA | 0.000010 | mg/L | <0.000010 | <0.000010 | <0.000010 | <0.000010 | ---- |
| **Thorium, total** | 7440-29-1 | E420/VA | 0.00010 | mg/L | <0.00010 | <0.00010 | <0.00010 | <0.00010 | ---- |
| **Tin, total** | 7440-31-5 | E420/VA | 0.00010 | mg/L | <0.00010 | <0.00010 | <0.00010 | <0.00010 | ---- |
| **Titanium, total** | 7440-32-6 | E420/VA | 0.00030 | mg/L | <0.00030 | <0.00030 | <0.00030 | <0.00030 | ---- |
| **Tungsten, total** | 7440-33-7 | E420/VA | 0.00010 | mg/L | <0.00010 | <0.00010 | <0.00010 | <0.00010 | ---- |
| **Uranium, total** | 7440-61-1 | E420/VA | 0.000010 | mg/L | <0.000010 | <0.000010 | <0.000010 | <0.000010 | ---- |
| **Vanadium, total** | 7440-62-2 | E420/VA | 0.00050 | mg/L | <0.00050 | <0.00050 | <0.00050 | <0.00050 | ---- |
| **Zinc, total** | 7440-66-6 | E420/VA | 0.0030 | mg/L | 0.0048 | <0.0030 | 0.0056 | 0.0141 | ---- |

## Analytical Results

| Sub-Matrix: Water (Matrix: Water) | | | | Client sample ID | FJCC | Town Hall | Fire Depatment | RCMP | ---- |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Client sampling date / time | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | 25-Feb-2025 11:00 | ---- |
| Analyte | CAS Number | Method/Lab | LOR | Unit | VA25A4115-001 | VA25A4115-002 | VA25A4115-003 | VA25A4115-004 | ---- |
| | | | | | Result | Result | Result | Result | ---- |
| **Total Metals** | | | | | | | | | |
| **Zirconium, total** | 7440-67-7 | E420/VA | 0.00020 | mg/L | <0.00020 | <0.00020 | <0.00020 | <0.00020 | ---- |

Please refer to the General Comments section for an explanation of any result qualifiers detected.

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# QUALITY CONTROL INTERPRETIVE REPORT

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A4115** | Page | : 1 of 7 |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith BC Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby, British Columbia Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment Plant - Quarterly DT Sampling | Date Samples Received | : 26-Feb-2025 11:25 |
| PO | : 10880 | Issue Date | : 05-Mar-2025 15:36 |
| C-O-C number | : ---- | | |
| Sampler | : ---- | | |
| Site | : RFP Tender No. 2022-IS-20 Extension | | |
| Quote number | : Town of Ladysmith Standing Offer_V2 | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

**This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.**

**Key**

**Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.**
**CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.**
**DQO: Data Quality Objective.**
**LOR: Limit of Reporting (detection limit).**
**RPD: Relative Percent Difference.**

## *Workorder Comments*

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

## *Summary of Outliers*

### *Outliers : Quality Control Samples*

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

### *Outliers: Reference Material (RM) Samples*

- No Reference Material (RM) Sample outliers occur.

## Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

## Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

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# CERTIFICATE OF ANALYSIS (GUIDELINE EVALUATION)

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A4112** | Page | : 1 of 4 |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith BC Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby, British Columbia Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment Plant - Weekly Sampling | Date Samples Received | : 26-Feb-2025 11:25 |
| PO | : 10880 | Date Analysis Commenced | : 26-Feb-2025 |
| C-O-C number | : ---- | Issue Date | : 03-Mar-2025 14:26 |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer_V2 | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Guideline Comparison

**Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).**

## *Signatories*

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Lindsay Gung | Supervisor - Water Chemistry | Inorganics, Burnaby, British Columbia |
| Lindsay Gung | Supervisor - Water Chemistry | Microbiology, Burnaby, British Columbia |
| Miles Gropen | Department Manager - Inorganics | Inorganics, Burnaby, British Columbia |

## No Breaches Found

## *General Comments*

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.
Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QA/QC Compliance Assessment to assist with Quality Review and Sample Receipt Notification.

When sampling time information is not provided by the client, sampling dates are shown without a time component. In these instances, the time component has been assumed by the laboratory for processing purposes.

Application of guidelines is provided "as is" without warranty of any kind, either expressed or implied, including, but not limited to fitness for a particular purpose, or non-infringement. ALS assumes no responsibility for errors or omissions in the information. Guidelines are not adjusted for the hardness, pH or temperature of the sample (the most conservative values are used). Measurement uncertainty is not applied to test results prior to comparison with specified criteria values.

Key : LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| µS/cm | microsiemens per centimetre |
| CFU/mL | colony forming units per millilitre |
| CU | colour units (1 cu = 1 mg/l pt) |
| mg/L | milligrams per litre |
| MPN/100mL | most probable number per hundred millilitres |
| NTU | nephelometric turbidity units |
| pH units | pH units |

>: greater than.
<: less than.
Red shading is applied where the result or the LOR is greater than the Guideline Upper Limit (or lower than the Guideline Lower Limit, if applicable).
For drinking water samples, Red shading is applied where the result for E.coli, fecal or total coliforms is greater than or equal to the Guideline Upper Limit.

## Workorder Comments

**Sample(s) 004: Exceeded Recommended Holding Time prior to receipt at the lab for HPC analysis.**

## Analytical Results Evaluation

Matrix: **Water**

| Analyte | CAS Number | Method/Lab | Unit | Client sample ID | Raw Water | DAF Effluent | UF Effluent | Treated Water (post reservoir) | ---- | ---- | ---- |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Sampling date/time | 25-Feb-2025 10:30 | 25-Feb-2025 10:30 | 25-Feb-2025 10:30 | 25-Feb-2025 10:30 | ---- | ---- | ---- |
| | | | | Sub-Matrix | Water | Water | Water | Water | ---- | ---- | ---- |
| | | | | | VA25A4112-001 | VA25A4112-002 | VA25A4112-003 | VA25A4112-004 | -------- | -------- | -------- |
| **Physical Tests** | | | | | | | | | | | |
| **Alkalinity, total (as CaCO3)** | ---- | E290/VA | mg/L | | ---- | ---- | ---- | 16.9 | ---- | ---- | ---- |
| **Colour, true** | ---- | E329/VA | CU | | ---- | ---- | ---- | <5.0 | ---- | ---- | ---- |
| **Conductivity** | ---- | E100/VA | µS/cm | | ---- | ---- | ---- | 68.1 | ---- | ---- | ---- |
| **pH** | ---- | E108/VA | pH units | | ---- | ---- | ---- | 7.31 | ---- | ---- | ---- |
| **Turbidity** | ---- | E121/VA | NTU | | ---- | ---- | ---- | <0.10 | ---- | ---- | ---- |
| **Organic / Inorganic Carbon** | | | | | | | | | | | |
| **Carbon, dissolved organic [DOC]** | ---- | E358-L/VA | mg/L | | 2.86 | 1.45 | 1.28 | ---- | ---- | ---- | ---- |
| **Carbon, total organic [TOC]** | ---- | E355-L/VA | mg/L | | 2.87 | 1.38 | 1.21 | ---- | ---- | ---- | ---- |
| **Microbiological Tests** | | | | | | | | | | | |
| **Heterotrophic plate count [HPC]** | ---- | E020/VA | CFU/mL | | ---- | ---- | ---- | <1 | ---- | ---- | ---- |
| **Coliforms, Escherichia coli [E. coli]** | ---- | E010/VA | MPN/100 mL | | ---- | ---- | ---- | <1 | ---- | ---- | ---- |
| **Coliforms, total** | ---- | E010/VA | MPN/10 0mL | | ---- | ---- | ---- | <1 | ---- | ---- | ---- |

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

## Summary of Guideline Limits

| Analyte | CAS Number | Unit | BCDWQG AO | BCDWQG MAC | BCDWQG OG | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Physical Tests** | | | | | | | | | |
| Alkalinity, total (as CaCO3) | ---- | mg/L | -- | -- | -- | | | | |
| Colour, true | ---- | CU | **15 CU** | -- | -- | | | | |
| Conductivity | ---- | µS/cm | -- | -- | -- | | | | |
| pH | ---- | pH units | -- | -- | **7 - 10.5 pH units** | | | | |
| Turbidity | ---- | NTU | -- | **1 NTU** | -- | | | | |
| **Organic / Inorganic Carbon** | | | | | | | | | |
| Carbon, dissolved organic [DOC] | ---- | mg/L | | | | | | | |
| Carbon, total organic [TOC] | ---- | mg/L | | | | | | | |
| **Microbiological Tests** | | | | | | | | | |
| Coliforms, Escherichia coli [E. coli] | ---- | MPN/100mL | -- | **1 MPN/100mL** | -- | | | | |
| Coliforms, total | ---- | MPN/100mL | -- | **1 MPN/100mL** | -- | | | | |
| Heterotrophic plate count [HPC] | ---- | CFU/mL | -- | -- | -- | | | | |

Please refer to the General Comments section for an explanation of any qualifiers detected.

**Key:**

| | |
|---|---|
| BCDWQG | British Columbia Drinking Water Quality Guidelines (JAN, 2023) |
| AO | Aesthetic Objective/Other Value |
| MAC | Maximium Acceptable Concentrations |
| OG | Operational Guidance |

# CERTIFICATE OF ANALYSIS

| | | | | |
|---|---|---|---|---|
| **Work Order** | : | **VA25A4112** | | |
| Client | : | **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : | Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : | 410 Esplanade PO Box 220<br>Ladysmith British Columbia Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby BC Canada V5A 1W9 |
| Telephone | : | ---- | Telephone | : +1 604 253 4188 |
| Project | : | Arbutus Water Treatment Plant - Weekly Sampling | Date Samples Received | : 26-Feb-2025 11:25 |
| PO | : | 10880 | Date Analysis Commenced | : 26-Feb-2025 |
| C-O-C number | : | ---- | Issue Date | : 03-Mar-2025 14:27 |
| Sampler | : | ---- | | |
| Site | : | Town of Ladysmith | | |
| Quote number | : | Town of Ladysmith Standing Offer | | |
| No. of samples received | : | 4 | | |
| No. of samples analysed | : | 4 | | |

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

## *Signatories*

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

| *Signatories* | *Position* | *Laboratory Department* |
|---|---|---|
| Lindsay Gung | Supervisor - Water Chemistry | Inorganics, Burnaby, British Columbia |
| Lindsay Gung | Supervisor - Water Chemistry | Microbiology, Burnaby, British Columbia |
| Miles Gropen | Department Manager - Inorganics | Inorganics, Burnaby, British Columbia |

## General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key: CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances.
LOR: Limit of Reporting (detection limit).

| *Unit* | *Description* |
|---|---|
| CFU/mL | colony forming units per millilitre |
| CU | colour units (1 cu = 1 mg/l pt) |
| mg/L | milligrams per litre |
| MPN/100mL | most probable number per hundred millilitres |
| NTU | nephelometric turbidity units |
| pH units | pH units |
| µS/cm | microsiemens per centimetre |

<: less than.
>: greater than.
Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.
UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

## Workorder Comments

**Sample(s) 004: Exceeded Recommended Holding Time prior to receipt at the lab for HPC analysis.**

## Analytical Results

**Sub-Matrix: Water**
**(Matrix: Water)**

| Analyte | CAS Number | Method/Lab | LOR | Unit | Raw Water | DAF Effluent | UF Effluent | Treated Water (post reservoir) | ---- |
|---|---|---|---|---|---|---|---|---|---|
| *Client sample ID* | | | | | | | | | |
| *Client sampling date / time* | | | | | 25-Feb-2025 10:30 | 25-Feb-2025 10:30 | 25-Feb-2025 10:30 | 25-Feb-2025 10:30 | ---- |
| | | | | | VA25A4112-001 | VA25A4112-002 | VA25A4112-003 | VA25A4112-004 | ---- |
| | | | | | Result | Result | Result | Result | ---- |
| **Physical Tests** | | | | | | | | | |
| **Alkalinity, total (as CaCO3)** | ---- | E290/VA | 1.0 | mg/L | ---- | ---- | ---- | 16.9 | ---- |
| **Colour, true** | ---- | E329/VA | 5.0 | CU | ---- | ---- | ---- | <5.0 | ---- |
| **Conductivity** | ---- | E100/VA | 2.0 | µS/cm | ---- | ---- | ---- | 68.1 | ---- |
| **pH** | ---- | E108/VA | 0.10 | pH units | ---- | ---- | ---- | 7.31 | ---- |
| **Turbidity** | ---- | E121/VA | 0.10 | NTU | ---- | ---- | ---- | <0.10 | ---- |
| **Organic / Inorganic Carbon** | | | | | | | | | |
| **Carbon, dissolved organic [DOC]** | ---- | E358-L/VA | 0.50 | mg/L | 2.86 | 1.45 | 1.28 | ---- | ---- |
| **Carbon, total organic [TOC]** | ---- | E355-L/VA | 0.50 | mg/L | 2.87 | 1.38 | 1.21 | ---- | ---- |
| **Microbiological Tests** | | | | | | | | | |
| **Heterotrophic plate count [HPC]** | ---- | E020/VA | 1 | CFU/mL | ---- | ---- | ---- | <1 | ---- |
| **Coliforms, Escherichia coli [E. coli]** | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |
| **Coliforms, total** | ---- | E010/VA | 1 | MPN/100 mL | ---- | ---- | ---- | <1 | ---- |

Please refer to the General Comments section for an explanation of any result qualifiers detected.

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# QUALITY CONTROL INTERPRETIVE REPORT

| | | | |
|---|---|---|---|
| **Work Order** | : **VA25A4112** | Page | : 1 of 7 |
| Client | : **Town of Ladysmith** | Laboratory | : ALS Environmental - Vancouver |
| Contact | : Shawn Baker | Account Manager | : Kevin Bhikadia |
| Address | : 410 Esplanade PO Box 220<br>Ladysmith BC Canada V9G 1A2 | Address | : 8081 Lougheed Highway<br>Burnaby, British Columbia Canada V5A 1W9 |
| Telephone | : ---- | Telephone | : +1 604 253 4188 |
| Project | : Arbutus Water Treatment Plant - Weekly Sampling | Date Samples Received | : 26-Feb-2025 11:25 |
| PO | : 10880 | Issue Date | : 03-Mar-2025 14:26 |
| C-O-C number | : ---- | | |
| Sampler | : ---- | | |
| Site | : Town of Ladysmith | | |
| Quote number | : Town of Ladysmith Standing Offer_V2 | | |
| No. of samples received | : 4 | | |
| No. of samples analysed | : 4 | | |

**This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.**

**Key**

**Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.**
**CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.**
**DQO: Data Quality Objective.**
**LOR: Limit of Reporting (detection limit).**
**RPD: Relative Percent Difference.**

## *Workorder Comments*

Holding times are displayed as "---" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

## *Summary of Outliers*

### *Outliers : Quality Control Samples*

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

### *Outliers: Reference Material (RM) Samples*

- No Reference Material (RM) Sample outliers occur.

## Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

## Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.